Chapter 3. Elementary Counting Problems

There Are a Lot of Them

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Pigeon-holes and induction draw conclusions *from* numbers; this chapter develops the basic techniques for *finding* them.

3.1 Permutations

Motivation. Arranging distinct objects in a row is the most basic counting problem: n choices for the first place, $n-1$ for the second, and so on. Once that is settled, repeated objects are handled by labelling them to make them distinct and then dividing out the labellings.

n patients at a dentist can be ordered in n ways for the first chair, then $n-1$ for the second (for *each* first choice), and so on, giving $n(n-1)\cdots 2\cdot 1$ orders.

Definition 3.1 (Permutation, factorial)

An arrangement of distinct objects in a linear order, using each exactly once, is a **permutation** of them. The number $n! := n(n-1)\cdots 2\cdot 1$ of permutations of n objects is n **factorial**.

Theorem 3.2

An n-element set has $n!$ permutations.

Proof. There are n choices for the first position, and for each of them $n-1$ for the second, …, and 1 for the last; the choices multiply. ∎

By convention $0! = 1$: then $n!\,m!$ correctly counts the ways for two rooms of n and m people to form lines, also when $n = 0$. For the size of $n!$ we quote without proof **Stirling's formula**

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n, \qquad \text{i.e.} \quad \lim_{n\to\infty} \frac{n!}{\sqrt{2\pi n}(n/e)^n} = 1. \tag{3.1}$$

Example 3.3

Flags with three horizontal stripes of the distinct colours red, white, green: $3! = 6$ (RWG, RGW, WRG, WGR, GWR, GRW).

Example 3.4

Five red, three yellow and two white flowers can be planted in a row in $A = \frac{10!}{5!\,3!\,2!} = 2520$ ways.
Solution. Label the flowers of each colour to make all ten distinct. Each of the A colour patterns can be labelled in $5!\,3!\,2!$ ways, and the labelled rows are the $10!$ permutations of ten distinct objects. Hence $A \cdot 5!\,3!\,2! = 10!$.

A collection with repetitions is a **multiset**; a linear order of all its elements is a **multiset permutation**.

Theorem 3.5 (Multiset permutations)

Let $a_1 + \cdots + a_k = n$ be nonnegative integers. A multiset with a_i objects of type i ($i \in [k]$) has

$$\frac{n!}{a_1!\,a_2!\cdots a_k!}$$

linear orders.

Proof [Supplement]. Let A be the number of linear orders. Attach labels $1, \dots, a_i$ to the objects of type i; this makes all n objects distinct. Every permutation of the n labelled objects arises exactly once from a pair (linear order of the multiset, labelling), and for a fixed linear order the labels of type i can be assigned in $a_i!$ ways, independently for different types. Hence $A \cdot a_1! \cdots a_k! = n!$ by Theorem 3.2. ■

3.2 Strings over a finite alphabet

Motivation. Words of length k over an alphabet count functions $[k] \to$ alphabet; forbidding repeated letters counts injections. Bijections let us transfer such counts to other problems, which is the most flexible tool of enumeration.

Theorem 3.6

There are n^k strings of length k over an n-element alphabet.

Proof. Each of the k positions can independently be filled in n ways (letters may repeat). ■

Example 3.7

There are $9 \cdot 10^{k-1}$ positive integers with k digits: of the 10^k strings over $\{0, \dots, 9\}$, exactly 10^{k-1} start with 0. (Or: 9 choices for the first digit, 10 for each other.)

The bijective method. To check that a ballroom holds as many men as women, ask them to form couples; if nobody is left over, the numbers agree.

Definition 3.8 (Bijection)

Let X, Y be finite sets and $f : X \to Y$ a function such that (1) $f(a) = f(b) \Rightarrow a = b$, and (2) every $y \in Y$ equals $f(x)$ for some $x \in X$. Then f is a **bijection** from X onto Y; equivalently, each $y \in Y$ has exactly one preimage.

Definition 3.9 (Injection, surjection)

A function satisfying (1) is **one-to-one** (**injective**); one satisfying (2) is **onto** (**surjective**).

Proposition 3.10

If there is a bijection from a finite set X onto a finite set Y, then $|X| = |Y|$.

Proof. A bijection partitions $X \cup Y$ into pairs $\{x, f(x)\}$ with one element from each set, every element lying in exactly one pair. If there are m pairs, then $|X| = |Y| = m$. ■

Example 3.11

An n-element set has 2^n subsets (without induction, cf. Theorem 2.4).
Solution. Map $B \subseteq [n]$ to the binary string whose i-th digit is 1 iff $i \in B$. Every binary string of length n comes from exactly one B, namely the set of positions of its 1's. By Theorem 3.6 there are 2^n such strings.

Example 3.12

Ten new intersections each get a traffic light or not, and lit ones may also get a gas station. Recording each intersection as A (light and gas station), B (light only) or C (neither) is a bijection onto words of length 10 over $\{A, B, C\}$, so there are 3^{10} possibilities.

Theorem 3.13

For $n \geq k$, the number of strings of length k over an n-element alphabet with no repeated letter is

$$(n)_k := n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}.$$

Proof. n choices for the first letter, $n-1$ for the second, …, $n-k+1$ for the k-th: the argument for $n!$, stopped after k steps. ■

Example 3.14

Filling five different cabinet posts from 20 candidates: $(20)_5 = 20 \cdot 19 \cdot 18 \cdot 17 \cdot 16$ ways.

3.3 Choice problems

Motivation. When only the set of chosen elements matters and not their order, we count subsets; every ordered selection of k elements corresponds to $k!$ orderings of the same subset. Shifting tricks then handle "no two consecutive" and "repetitions allowed".

How many tickets guarantee the jackpot of a lottery that draws 5 numbers from [90]? The order of the numbers does not matter.

Definition 3.15 (Binomial coefficient)

$\binom{n}{k}$ ("n choose k") denotes the number of k-element subsets of $[n]$; these are the **binomial coefficients** (Chapter 4).

Theorem 3.16

For $0 \leq k \leq n$: $\displaystyle\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{(n)_k}{k!}.$

Proof. By Theorem 3.13 there are $(n)_k$ strings of k distinct elements of $[n]$. Each k-element subset appears among them exactly $k!$ times, once for each ordering of its elements (Theorem 3.2). Hence $\binom{n}{k} \cdot k! = (n)_k$. ■

So the lottery needs $\binom{90}{5} = 43\,949\,268$ tickets.

Definition 3.17 (Complement)

For $S \subseteq [n]$, the **complement** S^c is the set of $i \in [n]$ with $i \notin S$.

Proposition 3.18

For $0 \le k \le n$:
(1) $\binom{n}{k} = \binom{n}{n-k}$;
(2) $\binom{n}{0} = \binom{n}{n} = 1$.

Proof. (1) $S \mapsto S^c$ maps k-subsets of $[n]$ to $(n-k)$-subsets, and every $(n-k)$-subset T is the image of exactly one S, namely T^c. Apply Proposition 3.10.

(2) The only 0-element subset is $\emptyset$; the other equality follows from (1). ■

In particular $\binom{0}{0} = 1$, and we set $\binom{n}{k} := 0$ for $k > n$.

Example 3.19

A student must work five days in January, never on two consecutive days. This can be done in $\binom{27}{5}$ ways.
Solution. For chosen days $a_1 < \cdots < a_5$ in $[31]$, "no two consecutive" is equivalent to $1 \le a_1 < a_2 - 1 < a_3 - 2 < a_4 - 3 < a_5 - 4 \le 27$. So $(a_i) \mapsto (a_i - (i-1))$ is a bijection onto the 5-element subsets of $[27]$.

Example 3.20

If the five lottery numbers from $[90]$ are drawn *with replacement* and only the multiset counts, $\binom{94}{5}$ tickets are needed.
Solution. Running the previous trick backwards, $f(b_1, \dots, b_5) = (b_1, b_2+1, b_3+2, b_4+3, b_5+4)$ maps multisets

$$1 \le b_1 \le b_2 \le b_3 \le b_4 \le b_5 \le 90 \tag{3.2}$$

bijectively onto the 5-element subsets of $[94]$: the b_i satisfy (3.2) iff the shifted values are strictly increasing in $[94]$.

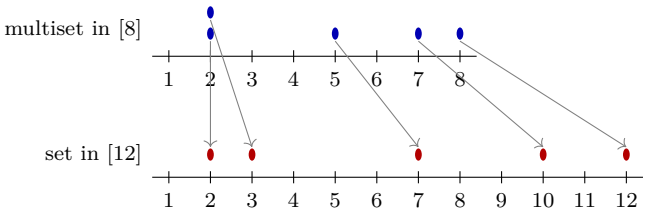


Figure. *Example 3.20 with smaller numbers: the multiset* $\{2,2,5,7,8\}$ *of* $[8]$ *is sent to the set* $\{2,3,7,10,12\}$ *of* $[12]$ *by adding* $0,1,2,3,4$*; equal entries become distinct.*

Theorem 3.21

The number of k-element multisets with elements from $[n]$ is $\binom{n+k-1}{k}$.

Proof [Supplement]. Write a multiset as $1 \le b_1 \le \cdots \le b_k \le n$ and let $f(b_1, \dots, b_k) := (b_1, b_2+1, \dots, b_k+k-1)$. The inequalities $b_i \le b_{i+1}$ are equivalent to $b_i + (i-1) < b_{i+1} + i$, and $b_k \le n$ to $b_k + k - 1 \le n + k - 1$. So f is a bijection onto the k-element subsets of $[n+k-1]$, with inverse subtracting $0, 1, \dots, k-1$ from the increasingly ordered elements. Conclude with Proposition 3.10. ■

	Parameters	Formula
Permutations	n distinct objects	$n!$
	a_i objects of type i, $\sum a_i = n$	$\frac{n!}{a_1!\cdots a_k!}$
Lists	k of n distinct objects	$(n)_k = \frac{n!}{(n-k)!}$
	words of length k over n letters	n^k
Subsets	k-element subsets of $[n]$	$\binom{n}{k}$
	k-element multisets from $[n]$	$\binom{n+k-1}{k}$

3.4 Selected exercises

Motivation. These book exercises show the two basic tools of the chapter at work: interpreting a quotient of factorials as a count, and building a bijection or recursive construction instead of computing.

Exercise 3.17 (Products of factorials)

Let $k \geq 1$ and let $b_1, \ldots, b_k$ be positive integers whose sum is less than the positive integer n. Prove that $b_1!\, b_2! \cdots b_k! < n!$. Can the statement be made stronger?

Solution.

Step 1 (a count). Let $s = b_1 + \cdots + b_k$, so $1 \leq s \leq n-1$. By Theorem 3.5, a multiset with b_i objects of type i has $\frac{s!}{b_1!\cdots b_k!}$ permutations. This number counts a nonempty set (for instance, all objects of type 1 first, then type 2, and so on), so it is a positive integer, and

$$b_1!\, b_2! \cdots b_k! \leq s!.$$

Step 2 (conclusion). Since $s \leq n-1$, $s! \leq (n-1)!$, and $(n-1)! < n!$ because $n \geq 2$. Hence $b_1! \cdots b_k! \leq (n-1)! = \frac{n!}{n} < n!$.

This is the stronger statement: the product is at most $\frac{n!}{n}$, and even at most $(b_1 + \cdots + b_k)!$, with equality iff $k = 1$ (for $k \geq 2$ two different types can be swapped, so there are at least two arrangements). ■

Exercise 3.26 (Permutations without peaks)

How many permutations of $[n]$ have no entry that is larger than both of its neighbours? (The first and the last entry have only one neighbour and never violate the condition.)

Solution. Call such permutations *peakless* and let a_n be their number. We show $a_n = 2^{n-1}$ for $n \geq 1$.

Step 1 (where n is). In a peakless permutation of $[n]$, $n \geq 2$, the entry n is larger than all its neighbours, so it has only one neighbour: n is the first or the last entry.

Step 2 (removing n). Deleting n from a peakless permutation of $[n]$ gives a peakless permutation of $[n-1]$: the entry next to n becomes an end entry, and every other entry keeps its neighbours.

Step 3 (inserting n). Conversely, let q be peakless on $[n-1]$ and put n at the front or at the end. The new entry n is an end entry. The entry next to it now has the neighbour n, which is larger, so it is not a peak; all other entries keep their neighbours. So both results are peakless, and they are different for $n \geq 2$.

Step 4 (counting). Steps 1–3 give a bijection between peakless permutations of $[n]$ and pairs $(q,$ side$)$, so $a_n = 2a_{n-1}$ for $n \geq 2$. With $a_1 = 1$ this gives $a_n = 2^{n-1}$. ■

Exercise 3.57 (Matrices with even lines)

How many $n \times n$ matrices with entries 0 and 1 have an even sum in each row and each column?

Solution. The answer is $2^{(n-1)^2}$. Call such matrices *even*; all congruences below are modulo 2.

Step 1 (the map). Send an even matrix M to its upper left $(n-1) \times (n-1)$ block B. There are $2^{(n-1)^2}$ possible blocks (Theorem 3.6), so it suffices to show that every B extends to exactly one even matrix.

Step 2 (forced entries). Given $B = (b_{ij})$, the row condition forces $m_{in} \equiv \sum_{j<n} b_{ij}$ for $i < n$, and the column condition forces $m_{nj} \equiv \sum_{i<n} b_{ij}$ for $j < n$. Each entry is 0 or 1, so these entries are uniquely determined.

Step 3 (the corner). Row n is even iff $m_{nn} \equiv \sum_{j<n} m_{nj} \equiv \sum_{i,j<n} b_{ij}$, and column n is even iff $m_{nn} \equiv \sum_{i<n} m_{in} \equiv \sum_{i,j<n} b_{ij}$. Both conditions require the same value, so exactly one choice of m_{nn} works.

Step 4 (conclusion). Every block extends to exactly one even matrix, so the map of Step 1 is a bijection and there are $2^{(n-1)^2}$ even matrices (Proposition 3.10). ■